SAINT MICHAEL COLLEGE OF CARAGA

PREDICTING STUDENT ACADEMIC STANDING USING MACHINE LEARNING MODELS

WITH EXPLAINABLE AI

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ABSTRACT

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(RESEARCH DESIGN)

I. OBJECTIVES

II. METHODOLOGY

III. FINDINGS

IV. RECOMMENDATIONS

KEYWORDS:

DEDICATION

This thesis is addressed to our beloved families, who have given us so much love, strength, and motivation throughout our journey.

To our parents, who have sacrificed so much for us to continue our studies, and towhom we've never stopped trying to live up to their belief in us, your unwavering belief and support in your children, and your prayers and encouragement keep us moving every day. This work is due to your guidance and endless support.

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CHAPTER 1

INTRODUCTION

1.1 Project Context/Background of the Study

In the rapidly advancing arena of educational research, Educational Data Mining (EDM) and Learning Analytics (LA) have proven transformative in enhancing institutional abilities to monitor and predict student performance. Traditional assessment paradigms centered around grades or instructor evaluations often fail to capture the multitude of dimensions affecting academic success. EDM and LA leverage emerging machine learning (ML) capabilities to provide higher education institutions with tools for predictive analytics and decision-making.

Machine learning algorithms such as Random Forest (RF), Support Vector Machines (SVM), Decision Trees (DT), are pivotal in analyzing multi-faceted educational datasets. While Random Forest is highly effective in handling nonlinear relationships, Decision Trees offer interpretative simplicity, especially in datasets with limited complexity. Rigorous comparative analyses of ML frameworks reveal that ensemble model like the Ant Colony Optimization-Decision Tree (ACO-DT) combination and RF outperform single algorithms in precision and early detection of at-risk students [3, 4].

Behavioral data integration serves to enhance the robustness of predictive systems. Incorporating metrics such as Learning Management System (LMS) activity, forum engagement, and task-specific time allocations can yield higher prediction accuracy while fostering adaptive learning environments. Research confirms that combining behavioral data with academic records (e.g., grades, attendance) yields actionable insights for personalized interventions. For example, one study utilized Moodle LMS data to provide real-time alerts for academic advising [5, 6].Localized research in contexts like the Philippines exemplifies the power of predictive models tailored to institutional environments. The application of Decision Trees to diploma programs demonstrated over 75% accuracy in predicting outcomes correlated with attendance and prior grades [7, 8]. National datasets such as PISA have been instrumental in enabling predictive analytics to address macro-level educational challenges. At the global level, studies from the UK illustrate the role of EDM in supporting individualized learning by creating interventions aligned with student-specific needs [9, 10].Interoperability in data integration is increasingly recognized as a critical component of successful predictive analytics frameworks. Efficient integration of institutional, national, and behavioral datasets ensures scalability while addressing the diversity of educational settings. Robust frameworks like those leveraging balanced datasets improve both prediction reliability and model equity [11, 12]. For example, handling underrepresented class imbalances within datasets significantly enhances the efficacy of Random Forest models, particularly for smaller-scale applications [5, 6].The growing integration of ensemble systems into educational data mining reflects a paradigm shift toward scalable, predictive analytics frameworks that balance performance

with interpretability. Personalized interventions are invaluable in not only improving academic outcomes but also shaping equitable, learner-centered experiences. Teacher insights add value by contextualizing predictive findings within practical academic realities, driving innovations in differentiated learning environments [13, 14, 15].

Findings from multi-modal studies suggest that integrating demographic, behavioral, and institutional data enables proactive decision-making for academic support and allocation of resources [16, 17, 18, 19, 20]. As machine learning technologies advance, the capacity to refine predictive frameworks grows, inviting future research into optimizing both algorithms and the educational systems they serve.

This work advances prior research by nominating five machine learning classifiers—Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes—for head-to-head performance comparison using XAI techniques. All student data used are limited to academic, enrollment, course load, library usage, online activity logs, and study habits questionnaire results.

The dataset aggregates records from 1st to 3rd year students enrolled in BSIT, BSCS, and BLIS programs (IT, CS, BLIS Departments), with all selected students identified as face-to-face (F2F) enrollees. The aggregated features include academic grades, course load (number of units), library usage logs, online LMS engagement, study habits questionnaire responses, and other institutional indicators.

1.2 Literature Review

This chapter presents a comprehensive review of current research and technological developments related to predictive analytics in education, specifically focusing on the use of machine learning algorithms such as Random Forest to forecast student academic standing. The discussion explores how Educational Data Mining (EDM) and Learning Analytics (LA) have evolved to support early identification of at-risk students and enable data-driven decision-making in higher education. It further examines the effectiveness of various machine learning approaches in academic prediction, emphasizing the advantages of ensemble models like Random Forest for handling complex and multidimensional datasets. The review is organized thematically into seven major sections: (1) Educational Data Mining and Learning Analytics, (2) Machine Learning and Student Performance Prediction, (3) Application of the Random Forest Algorithm in Education, (4) Feature Selection and Data Preprocessing, (5) Model Evaluation Metrics, (6) Explainable Artificial Intelligence (XAI), and (7) Predictive Analytics in the Philippine Context. The final section synthesizes the findings, identifies existing research gaps, and establishes the foundation for developing a Random Forest–based predictive model tailored to the needs of Saint Michael College of Caraga (SMCC).

EDUCATIONAL DATA MINING AND LEARNING ANALYTICS

In recent years, higher education institutions have increasingly adopted **Educational Data Mining (EDM)** and **Learning Analytics (LA)** to enhance understanding of student performance and engagement. Traditional assessment methods, such as grades or instructor observations, provide limited insights into student learning behaviors. EDM and LA leverage data from Learning Management Systems (LMS), discussion forums, and assessments to enable data-driven decision-making and proactive interventions [1]–[5].

Aldowah et al. [1] highlight the role of EDM and LA in improving learning outcomes and institutional decision-making. Zheng and Li [3] emphasize the use of EDM in STEM education, predicting student disengagement and optimizing instructional strategies. Daniela and Mirela [2] discuss the integration of student activity logs to monitor learning behaviors, while Pektaş et al. [4] propose machine learning models for detecting misconceptions and personalizing interventions. Angeioplastis et al. [5] focus on leveraging LMS data with predictive algorithms to anticipate student performance.

II. Machine Learning in Student Performance Prediction

Machine learning models are central to predictive analytics in education, with Random Forest (RF), Gradient Boosting, Support Vector Machines (SVM), and Decision Trees among the most widely applied techniques [6]–[10]. These algorithms identify patterns in academic, behavioral, and demographic datasets, enabling early detection of students at risk of poor performance or dropout.

Al-Ameri et al. [6] demonstrate the effectiveness of ensemble models, including RF, in predicting academic outcomes from multimedia data. Sawarkar et al. [7] combine RF with Logistic Regression to improve accuracy in identifying at-risk learners. Borna et al. [8] analyze clickstream data to enhance prediction reliability, while Vijayalakshmi and Srinath [10] examine the impact of collaboration metrics on learning performance. Muhammad et al. [9] highlight personalized education datasets to support adaptive learning strategies.

APPLICATION MACHINE LEARNING CLASSIFICATION ALGORITHM MODELS

Classification algorithms play a fundamental role in data-driven research, with their performance shaped by factors such as the choice of algorithm, preprocessing, and evaluation metrics. This empirical study systematically compares Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes on benchmark datasets, finding that Random Forest consistently demonstrates the highest accuracy, precision, recall, and F1-score. The results indicate that no single evaluation metric is universally optimal; rather, a combination of metrics better captures the varied aspects of classification performance. These findings underscore the importance of careful metric and algorithm selection, while also suggesting the necessity of contextualizing classifier performance to the requirements of specific applications .

FEATURE SELECTION AND DATA PREPROCESSING

Data preprocessing and feature engineering significantly affect model performance. Techniques such as **normalization, SMOTE for class imbalance, and categorical encoding** are widely adopted to improve predictive accuracy [16]–[20].Angeioplastis et al. [16] emphasize preprocessing to handle imbalanced datasets. Vijayalakshmi and Srinath [17] highlight the integration of demographic and behavioral features. Pektaş et al. [18] and Chauhan and Mishra [19] demonstrate systematic feature engineering to improve classification outcomes, while Gul et al. [20] discuss preprocessing in the context of deep learning models for academic prediction.

MODEL EVALUATION METRICS

Evaluating predictive models requires standardized metrics to assess accuracy and reliability. Accuracy, precision, recall, and F1-score are frequently used to quantify model performance in educational prediction tasks [6], [21]–[24]. Al-Ameri et al. [6] emphasize the use of balanced metrics to evaluate RF performance, especially in imbalanced datasets. Sawarkar et al. [7] and Madahana & Ekoru [22] demonstrate comparative evaluation with Gradient Boosting and Decision Trees. Angeioplastis et al. [23] provide guidelines for cross- validation and stratified sampling to ensure generalizability, while Borna et al. [24] incorporate probabilistic metrics to assess prediction confidence.

EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

Despite their predictive power, machine learning models are often criticized as "black boxes." **Explainable AI (XAI)** techniques such as SHAP and LIME help educators interpret model predictions and trust intervention strategies [25]–[28]. Borna et al. [25] integrate LIME for visualization of feature importance in student performance prediction. Aldowah et al. [26] propose XAI frameworks in LA systems for transparency. Pektaş et al. [27] and Chauhan & Mishra [28] illustrate how interpretable insights can guide interventions, ensuring fairness and accountability.

VII. Predictive Analytics in the Philippine Context

Localized applications are critical for small-scale institutions like Saint Michael College of Caraga (SMCC). Limited data availability and specific demographic characteristics require customized predictive models [29], [30]. Muhammad et al. [29] provide structured datasets supporting RF-based prediction in Philippine institutions, while Aldowah et al. [30] highlight challenges such as sparse datasets and resource constraints.

1.3 Project Objectives

General Objective

To develop a predictive system that uses multiple machine learning models to classify student academic standing, providing interpretable results and actionable insights for educational decision-making.

Specific Objectives

1.To analyze requirements and identify the necessary data sources, system functions, and user needs for the predictive model.

2.To design a functional system that integrates institutional datasets for automated academic standing classification.

3.To implement and compare multiple machine learning models, including Decision Tree, Random Forest, SVM, KNN, and Naive Bayes, using a common training and testing dataset.

4.To evaluate the performance of each model using accuracy, precision, recall, F1-score, and ROC-AUC metrics.

5.To select the best-performing model and integrate it into the deployed system.

6.To assess the system's functionality, performance efficiency, usability, and reliability based on ISO 25010 standards to ensure software quality.

7.To incorporate Explainable Artificial Intelligence (XAI) techniques for model interpretability and transparency.

1.4 Scope and Limitations

This study focuses on the development of a predictive analytics system that employs the Random Forest algorithm to forecast student academic standing at Saint Michael College of Caraga, particularly within the College of Computing and Information Sciences (CCIS). The dataset includes third-year students enrolled in BSIT, BSCS, and BLIS programs. The model will classify students as Good Standing, On Probation, or At-Risk, based on variables such as

academic performance, attendance, demographics, and socioeconomic data. The predictive results will be presented through a decision-support dashboard for administrators and faculty to aid in early academic intervention and retention planning.

The model is limited to datasets collected from SMCC, which may affect generalizability to other institutions. Incomplete, inconsistent, or biased data may influence prediction accuracy. The system does not include non-academic factors such as mental health, employment status, or internet access, which also affect performance. Ethical considerations—such as data privacy, bias, and potential misclassification—are recognized, and measures will be taken to ensure confidentiality and responsible use. The prototype does not yet support long-term tracking or qualitative validation, and results are intended to assist, not replace, academic judgment.

1.5. Definition of Terms

In this study, the following terms are defined operationally based on their application in developing and evaluating a Random Forest-based predictive model for student academic performance at Saint Michael College of Caraga:

Academic Standing - Refers to the classification of students at Saint Michael College of Caraga based on their overall academic performance. In this study, students are categorized as *Good Standing*, *On Probation*, or *At-Risk* according to the predictive output of the Random Forest model.

Accuracy - The share of predictions the Random Forest model gets right out of all predictions made, giving a general sense of how well the classifier identifies students' academic standing overall. While useful as a headline metric, it is best interpreted alongside precision, recall, and the confusion matrix for a more comprehensive performance evaluation.

At-Risk - A classification output from the predictive model indicating students with a high likelihood of academic failure or dropout. These students are prioritized for academic counseling and targeted intervention based on their predicted risk level.

Confusion Matrix - A tabular representation used to evaluate the model's performance by comparing predicted outcomes with actual results. It shows how many instances were correctly and incorrectly classified for each category of academic standing.

Data Cleaning - A crucial step in data preprocessing that involves detecting, correcting, or removing inaccurate, incomplete, or inconsistent data entries from the dataset to ensure high-quality and reliable model training results.

Data Preprocessing - The process of cleaning, transforming, and organizing raw institutional data to prepare it for model training. This includes handling missing values, encoding categorical variables, and balancing class distributions to improve prediction accuracy.

Decision-Support System - The functional component of the study that presents predictive results through an interactive dashboard. It enables faculty and administrators to visualize model outcomes, monitor student performance, and make timely, data-driven academic decisions.

Educational Data Mining (EDM) - The process of applying data mining techniques to educational datasets to uncover hidden patterns and trends. In this study, EDM is used to identify academic, demographic, and socioeconomic factors that affect student performance and academic standing.

Explainable Artificial Intelligence (XAI) - A framework of methods and tools that make the predictions of machine learning models understandable to humans. The study integrates XAI to help educators interpret the model's reasoning behind each academic standing classification, ensuring transparency and trust.

F1-Score - A model evaluation metric that combines precision and recall into a single measure, representing their harmonic mean. It helps determine the balance between correctly identifying at-risk students and minimizing false classifications.

Feature Selection - The process of determining which input variables contribute most significantly to predicting student academic performance. It enhances model interpretability and reduces computational complexity by eliminating irrelevant or redundant features.

Good Standing - A predictive label assigned to students who meet or exceed institutional academic standards. Students in this category exhibit consistent performance and require minimal intervention from academic advisers.

Learning Analytics (LA) - The measurement and analysis of student learning behaviors using data-driven techniques. Within this project, LA supports the early identification of performance trends and assists in developing strategies for improving academic outcomes.

Machine Learning (ML) - A subset of artificial intelligence that enables computers to learn patterns from data without being explicitly programmed. This study applies ML techniques, specifically the Random Forest algorithm, to predict students' academic standing based on historical and institutional data.

Precision - A metric that measures the proportion of correctly predicted instances among all instances predicted for a specific category. In this study, it shows how accurately the model identifies students as Good Standing, On Probation, or At-Risk without false positives.

Predictive Analytics - The overarching framework of this study, involving statistical and machine learning methods to forecast student academic standing. It enables the institution to identify potential academic risks and implement early interventions.

Recall - A model evaluation metric that measures the proportion of actual instances of a class (for example, all At-Risk students) that the model correctly identifies. It reflects the model's sensitivity in detecting students who truly require academic intervention.

ROC-AUC (Receiver Operating Characteristic – Area Under the Curve) - A performance metric that evaluates the model's ability to distinguish between different classes. A higher ROC-AUC value indicates a stronger predictive capability in classifying students accurately across all categories.

SMOTE (Synthetic Minority Oversampling Technique) - A data-balancing method used during preprocessing to address class imbalance by generating synthetic examples of underrepresented categories, such as the At-Risk class, to improve model fairness and accuracy.

Student Retention - The institutional objective of keeping students enrolled and progressing toward graduation. The predictive model contributes to this goal by identifying students at risk of academic failure early, allowing for proactive support measures.

CHAPTER 2

2.1 Research Design

This study employs a quantitative, developmental research design. The quantitative approach is essential for the statistical analysis of student data, which includes academic, demographic, and socioeconomic variables. This design allows for the application of machine learning techniques to identify patterns and build a predictive model. The developmental aspect of the research involves the systematic design, creation, and evaluation of the predictive system as a software artifact.

2.2 Software Development Life Cycle

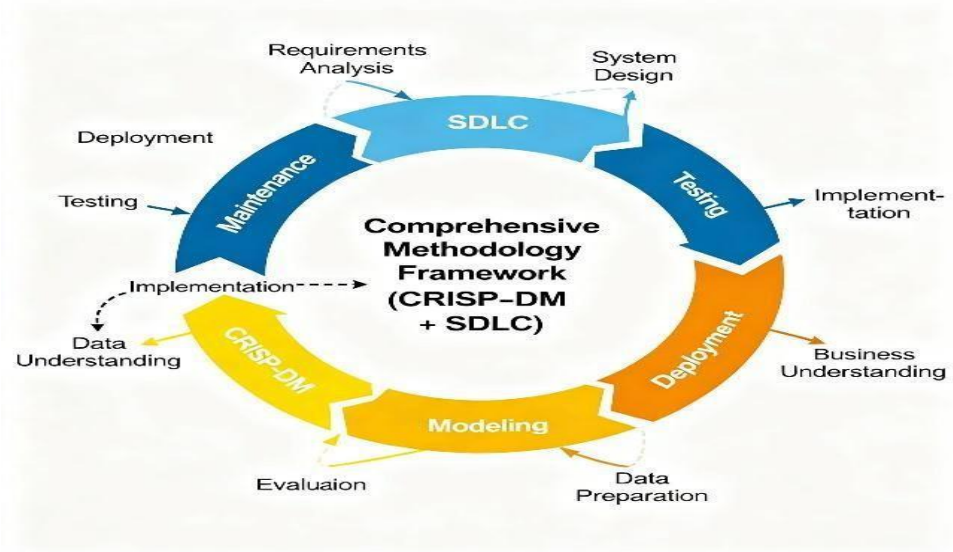


Figure 1: Comprehensive Methodology

This study employs a **comprehensive methodology** that synergizes the structured phases of an iterative **Software Development Life Cycle (SDLC)** with the cyclical, data-centric phases of the **Cross-Industry Standard Process for Data Mining (CRISP-DM)**. This integrated framework provides a robust structure for managing both the software engineering and data science components of the project. The SDLC governs the overall project architecture, from requirements analysis to final system deployment, while the CRISP-DM framework is nested within it to guide the iterative processes of data understanding, preparation, modeling, and evaluation. This dual approach ensures that the development of the predictive model is both methodologically sound and aligned with the software implementation goals.

2.3 System Architecture

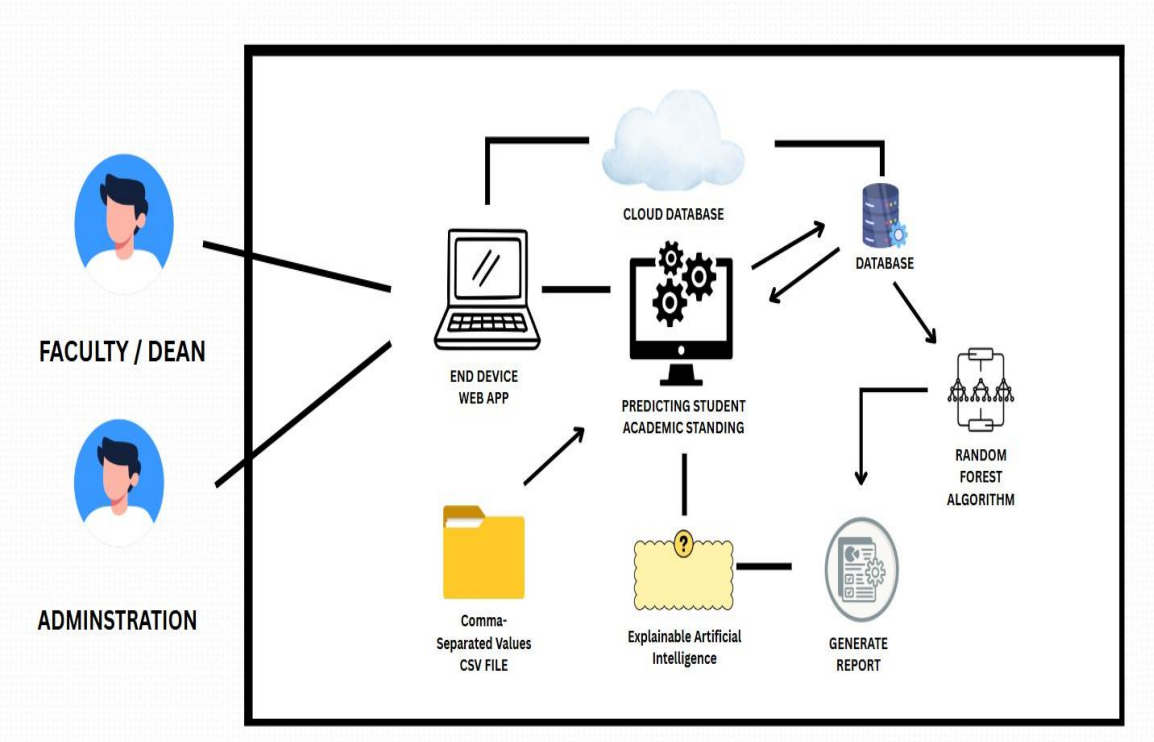


Figure 2:System Architecture

Figure 2:Illustrates the system is designed using a multi-tier web architecture, a standard paradigm for developing robust and scalable web-based applications. This architecture logically separates the system into three distinct layers: the Client-Side, the Server-Side, and the Data Tier. This separation ensures modularity, simplifies maintenance, and enhances security by isolating the business logic and data storage from the user interface.

2.4 Conceptual Framework

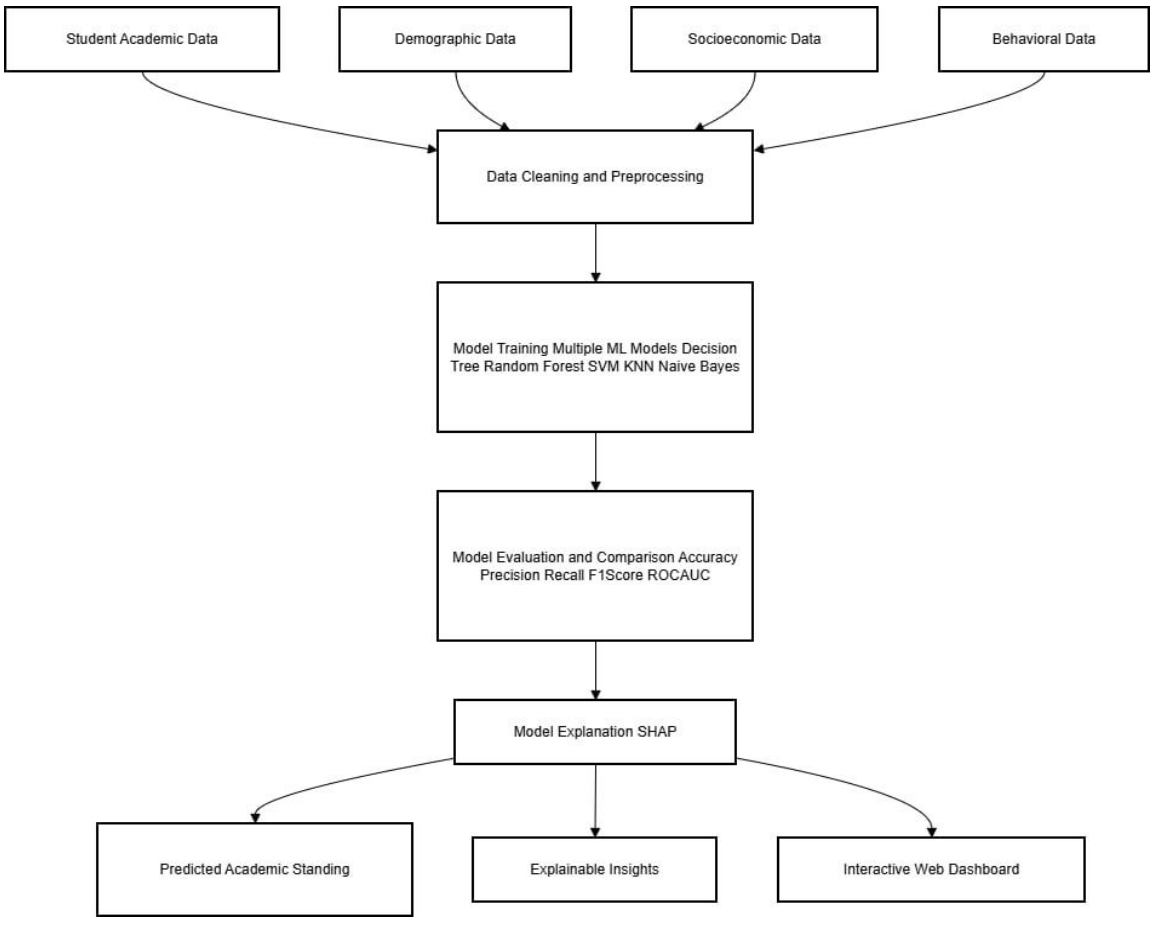


Figure 3: Conceptual Framework

Figure 3 the conceptual framework of this study is illustrated using an Input-Process- Output (IPO) model. This model provides a clear, high-level overview of the research flow, demonstrating how raw data is systematically transformed into actionable intelligence for educational stakeholders. The framework is composed of three sequential stages: the Input, the Process, and the Output.

In the process stage, the cleaned data are used to train and compare multiple machine learning algorithms, and the best-performing model is selected to generate academic standing predictions and explainable insights for stakeholders.

2.5 Algorithms / Framework Used

This study utilizes multiple machine learning algorithms to predict students' academic standing. These algorithms enable comparative analysis and selection of the most effective predictive model for educational data. The framework integrates (1) data-driven academic factors, (2) model training and prediction, and (3) measurability through standard classification performance metrics to ensure reliability of results.

Machine Learning Framework and System Workflow

The system follows a supervised learning classification framework where measured academic factors are transformed into features used to predict the student's academic standing category.

System Workflow:

Data Collection (Academic Factors Input)

Student academic factors (e.g., grades/GPA, attendance, course load/units, and other available indicators in the dataset) are gathered from institutional sources. Each factor is converted into measurable values such as averages, counts, or rates.

Preprocessing and Feature Preparation

Data are cleaned to handle missing or inconsistent records. Categorical variables (if present) are encoded, and numerical variables may be scaled/normalized as needed—especially for algorithms such as SVM and KNN. The output of this stage is a feature vector x representing each student.

Model Training and Validation

The dataset is split into training and testing sets (or validated through cross-validation). The following machine learning algorithms are trained using the same feature set: Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naive Bayes.

Prediction and Classification Output

For every student record, the trained model generates a predicted academic standing class Ĉ. The system outputs the prediction for decision support in identifying students who may require intervention.

Performance Evaluation (Measurability)

The system produces a confusion matrix and computes standard evaluation metrics to measure how effectively each model predicts each academic standing category. These metrics guide the selection of the best model for deployment.

Algorithms Used

Decision Tree Algorithm

The Decision Tree algorithm is employed due to its interpretability and ability to model non-linear relationships. It operates by recursively partitioning the feature space into decision nodes based on attribute values that minimize an impurity measure. For classification tasks, impurity is commonly measured using Gini impurity, calculated as:

$G = 1 - \Sigma (p_k^2)$, for k = 1 to K

where p_k represents the proportion of class k in a node and K denotes the total number of classes. Decision Trees provide transparent decision rules that aid in understanding how student attributes influence academic outcomes.

Role in the system: Generates human-readable IF–THEN rules that support explainability in academic decision-making.

Random Forest Algorithm

This study utilizes the Random Forest algorithm as the primary predictive model due to its high accuracy, robustness to overfitting, and ability to handle complex and multidimensional educational datasets. Random Forest operates by constructing an ensemble of decision trees (T_1, T_2, …, T_B), each trained on bootstrapped samples of the training data with random feature selection at each split to increase diversity.

For regression tasks, the predicted outcome $\hat{y}$ for an input vector x is computed as the mean prediction across all trees:

$$\hat{y} = (1/M) \Sigma T_j(x), \text{ for } j = 1 \text{ to } M$$

For classification tasks, the predicted class $\hat{C}$ is obtained through majority voting of individual tree predictions:

$$\hat{C} = \text{mode } \{T_1(x), T_2(x), \ldots, T_B(x)\}$$

This ensemble learning strategy improves generalization performance and predictive stability, making Random Forest well suited for educational prediction tasks.

Role in the system: Produces stable predictions using multiple trees and supports ranking of influential academic factors.

Support Vector Machine (SVM) Algorithm

The Support Vector Machine algorithm is applied due to its effectiveness in high-dimensional classification problems. SVM identifies an optimal decision boundary that maximally separates data points of different academic standing categories. The algorithm focuses on support vectors, which are the data points closest to the boundary, enhancing robustness and generalization in complex student datasets.

Role in the system: Separates academic standing categories using a margin-maximizing boundary; feature scaling is typically applied to ensure fair contribution of all academic factors.

K-Nearest Neighbors (KNN) Algorithm

The K-Nearest Neighbors algorithm is included due to its simplicity and effectiveness in pattern recognition. KNN classifies a student by identifying the k closest instances in the dataset using a distance-based measure. The predicted academic standing is determined by majority voting among the nearest neighbors. This approach enables the model to capture similarities among students with comparable academic and behavioral characteristics.

Role in the system: Predicts academic standing by comparing a student's factor profile to similar historical cases; scaling/normalization is often applied for fair distance computation.

Naïve Bayes Algorithm

The Naive Bayes algorithm is utilized due to its computational efficiency and probabilistic foundation. It estimates the probability of a student belonging to a particular academic category based on observed features and prior probabilities. Although it assumes feature independence, Naive Bayes performs effectively in many educational classification tasks and serves as a reliable baseline model.

Role in the system: Provides probability-based predictions and functions as a benchmark model for comparison.

Explainable Artificial Intelligence (XAI) Framework

Explainable Artificial Intelligence (XAI) is a framework designed to make machine learning models more transparent and interpretable. It encompasses a variety of algorithms, such as SHAP and LIME, that quantify and visualize how individual features influence model predictions. In this study, SHAP is integrated as the primary XAI method for explaining Random Forest outputs.

To enhance interpretability and stakeholder trust, the SHAP (SHapley Additive exPlanations) framework is incorporated to decompose individual model predictions into additive feature contributions ϕ_i. SHAP values are grounded in cooperative game theory, providing a mathematically principled method for attributing the effect of each feature.

The prediction model f(x) for an instance x can be expressed as:

$$f(x) = E[f(X)] + \Sigma_{i=1}^{M} \phi_i$$

where E[f(X)] is the expected output of the model over the dataset and M is the total number of input features. The SHAP values ϕ_i quantify each feature's marginal contribution averaged over all possible feature combinations. SHAP enables clear, consistent explanations of complex model outputs, supporting actionable insights for targeted academic interventions.

Model Evaluation Metrics

The model's predictive performance is assessed using standard classification metrics defined as follows:

Accuracy measures the proportion of correct predictions:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision assesses the correctness of positive predictions:

Precision = TP / (TP + FP)

Recall (Sensitivity) gauges the completeness of positive predictions:

Recall = TP / (TP + FN)

F1-Score provides a harmonic mean of precision and recall balancing false positives and false negatives:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Here, TP, TN, FP, and FN signify true positives, true negatives, false positives, and false negatives, respectively. Additionally, the area under the Receiver Operating Characteristic curve (ROC-AUC) is employed to evaluate the model's discrimination capacity across classification thresholds.

For multi-class academic standing, TP, TN, FP, and FN are computed per class using a one-vs-rest approach; precision, recall, and F1-score are reported per class and summarized using macro-averaged or weighted-averaged scores, and ROC-AUC is computed using a one-vs-rest strategy across classes.

In this study, Recall and F1-score for the At-Risk class are emphasized to minimize false negatives and support early academic intervention.

2.6 Use case Diagram

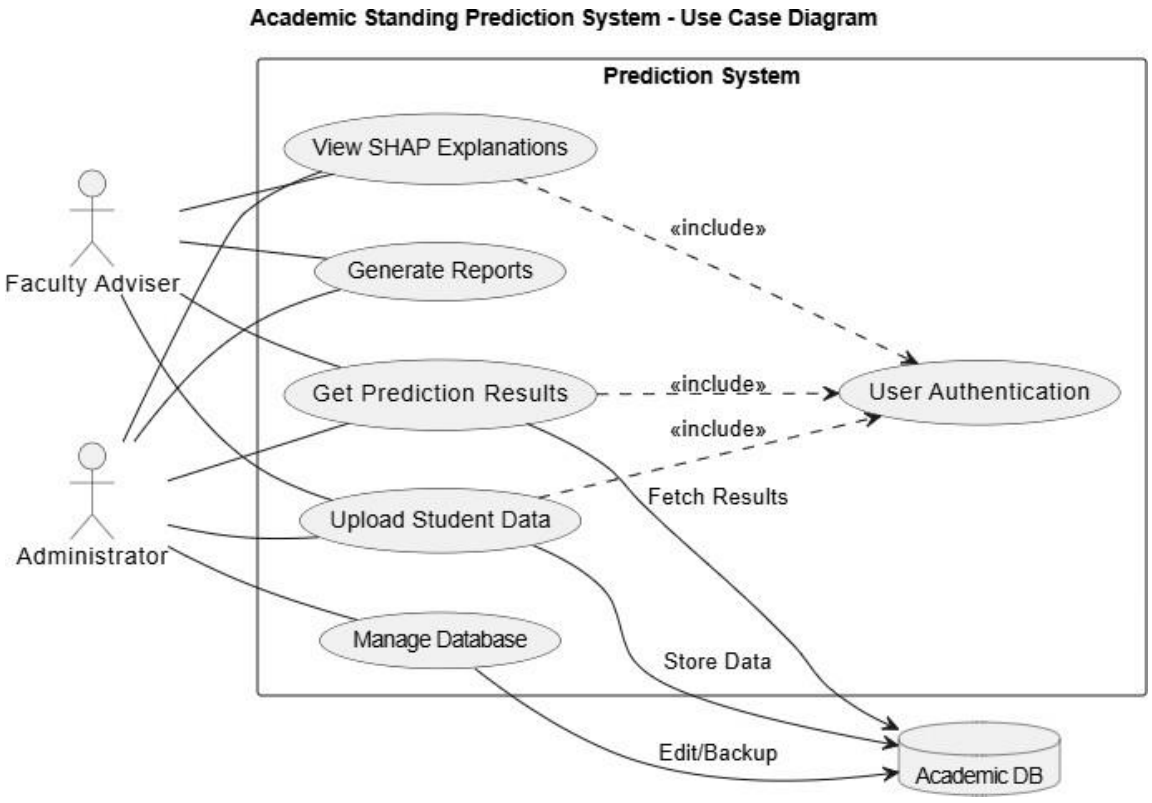


Figure 4: Use case Diagram

Figure 4 illustrates how administrators and faculty advisers interact with the system's core functions, with user authentication securing all critical actions and database interactions ensuring centralized and regulated data management.

2.7 Activity Diagram

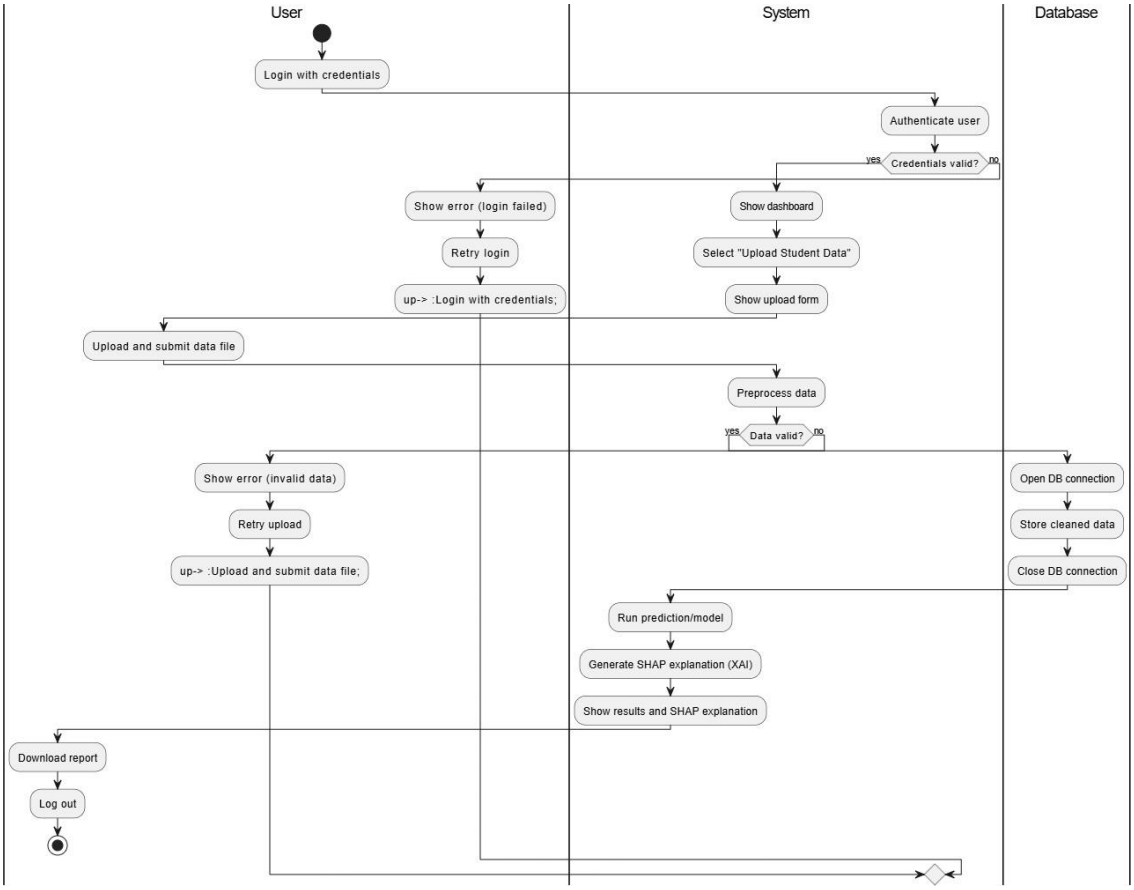


Figure 5: Activity Diagram

Figure 5 illustrates how administrators and faculty advisers interact with the prediction system's main workflow, including secure login, data upload, validation, and centralized database operations. After generating prediction results, the system produces SHAP-based explanations, ensuring users receive both outcomes and model reasoning. Decision nodes manage feedback and retries, while swimlanes delineate the responsibilities of users, system, and database throughout the process. The inclusion of SHAP explanations

enhances transparency and interpretability for all end users

2.8 Sequence Diagram

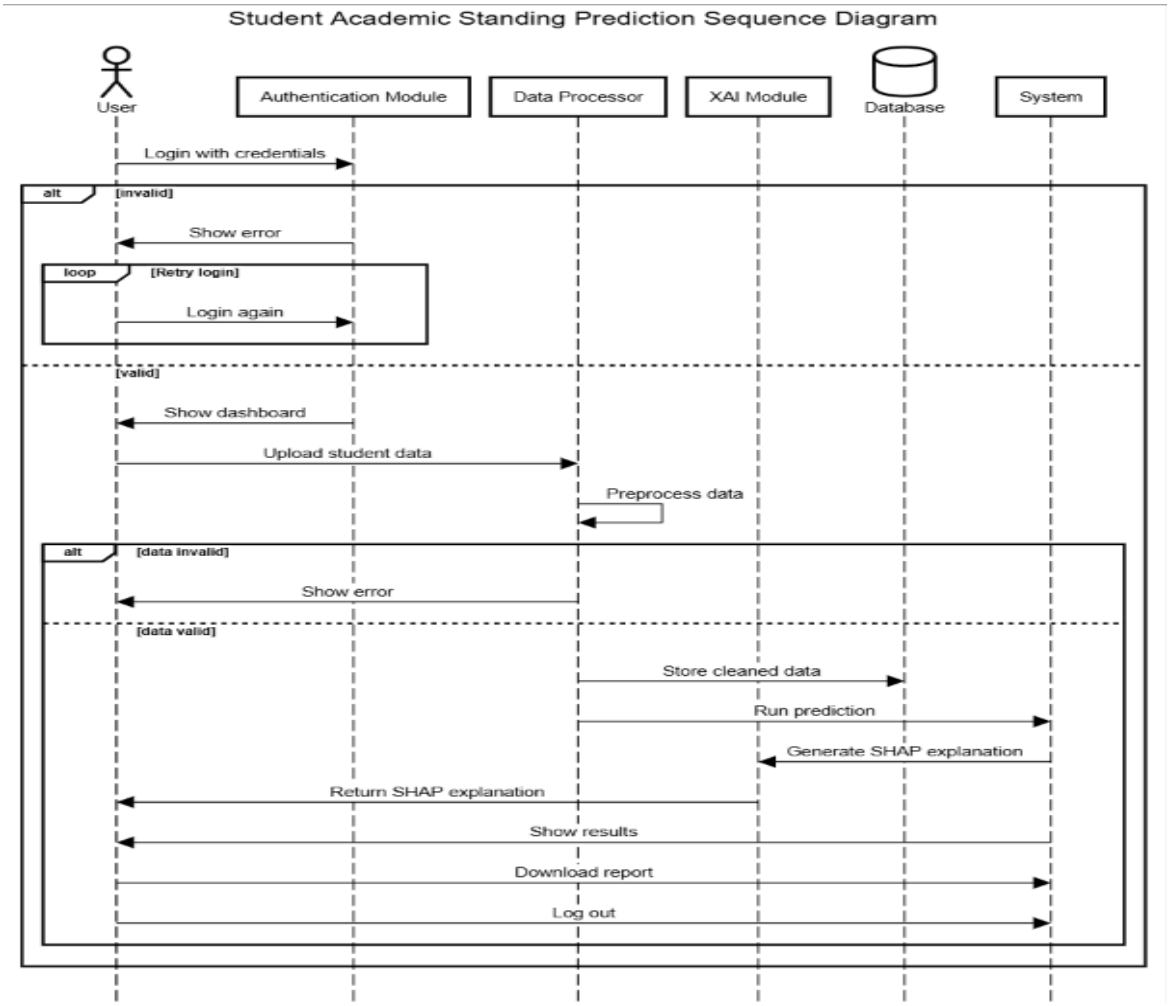


Figure 6: Sequence Diagram

Figure 6 shows how users interact with the prediction system to log in, upload and validate student data, and store it in the centralized database. The system then generates prediction results along with SHAP explanations, providing both outcomes and model reasoning. Conditional branches handle errors and retries, while distinct lifelines represent the roles of users, system modules, and the database. SHAP explanations ensure transparency and interpretability of the predictions.

2.9 ERD (Database Design)

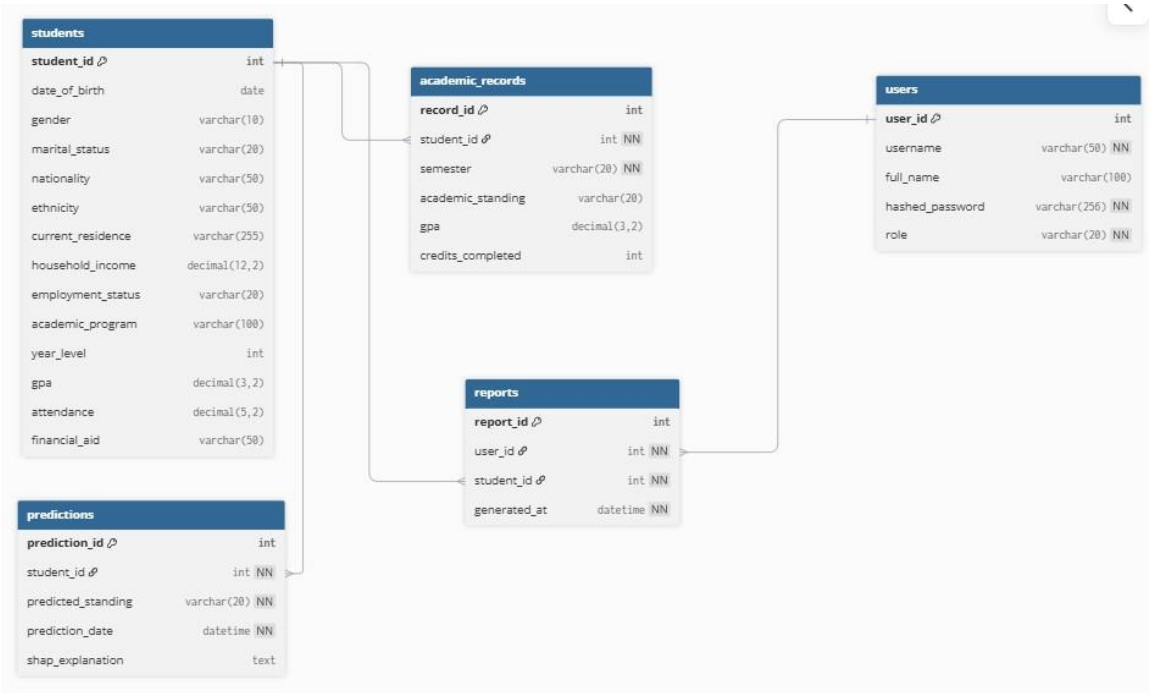


Figure 7: Entity Relationship Diagram

Figure 7 shows the core tables and relationships for managing students, academic records, predictions, user accounts, and reports. Each student is linked to their records and prediction results, while users can generate reports. This design ensures organized, secure data access for prediction and reporting in the academic system.

2.10 User Interface Design (Prototype)

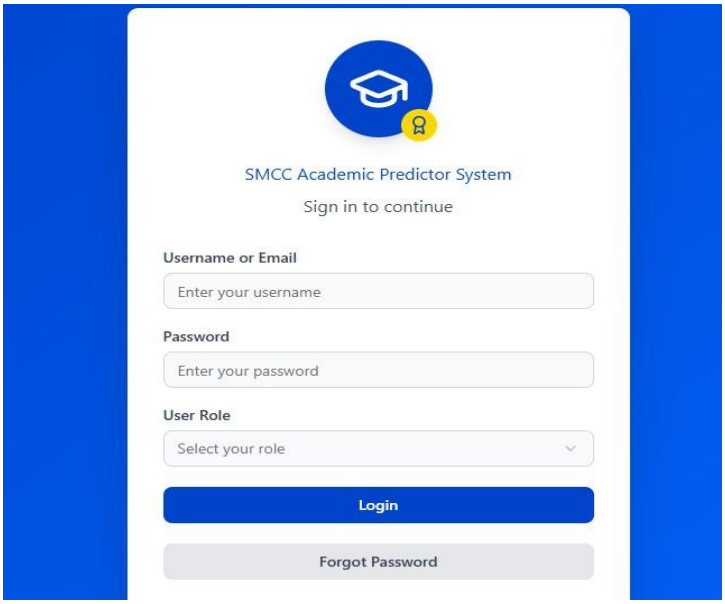


Figure 8: Login Screen

Figure 8 shows how users interact with the prediction system to log in, select their role, and authenticate access to the centralized application. The login form establishes user identity as the gateway to uploading data, generating predictions, and viewing model explanations. Conditional branches support error handling for invalid credentials, while clear lifelines distinguish authorized user roles and secure access, ensuring interpretability and integrity for all subsequent processes.

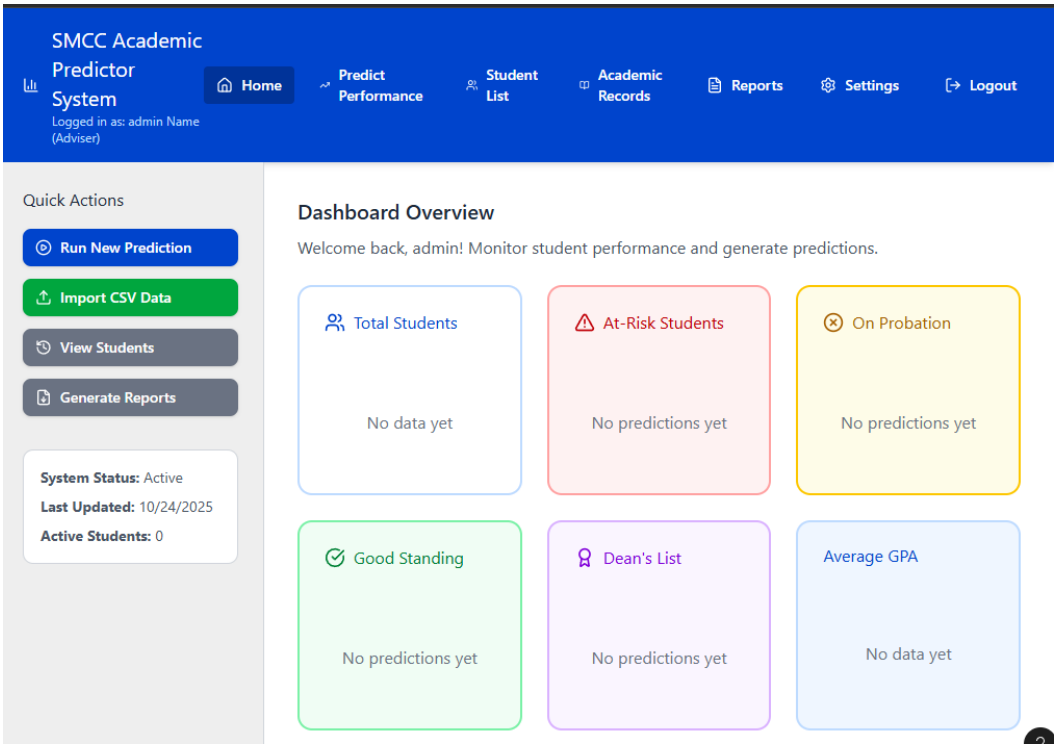


Figure 9: Dashboard/Home

This figure 9 shows how users interact with the prediction system dashboard to view summary metrics, monitor prediction outcomes, and access core features such as running new predictions or importing CSV data. Each dashboard card and quick action links users to data upload, model execution, and result validation. Conditional branches handle empty-state messaging and errors, while key indicators and SHAP-driven output provide a transparent, interpretable overview of prediction results.

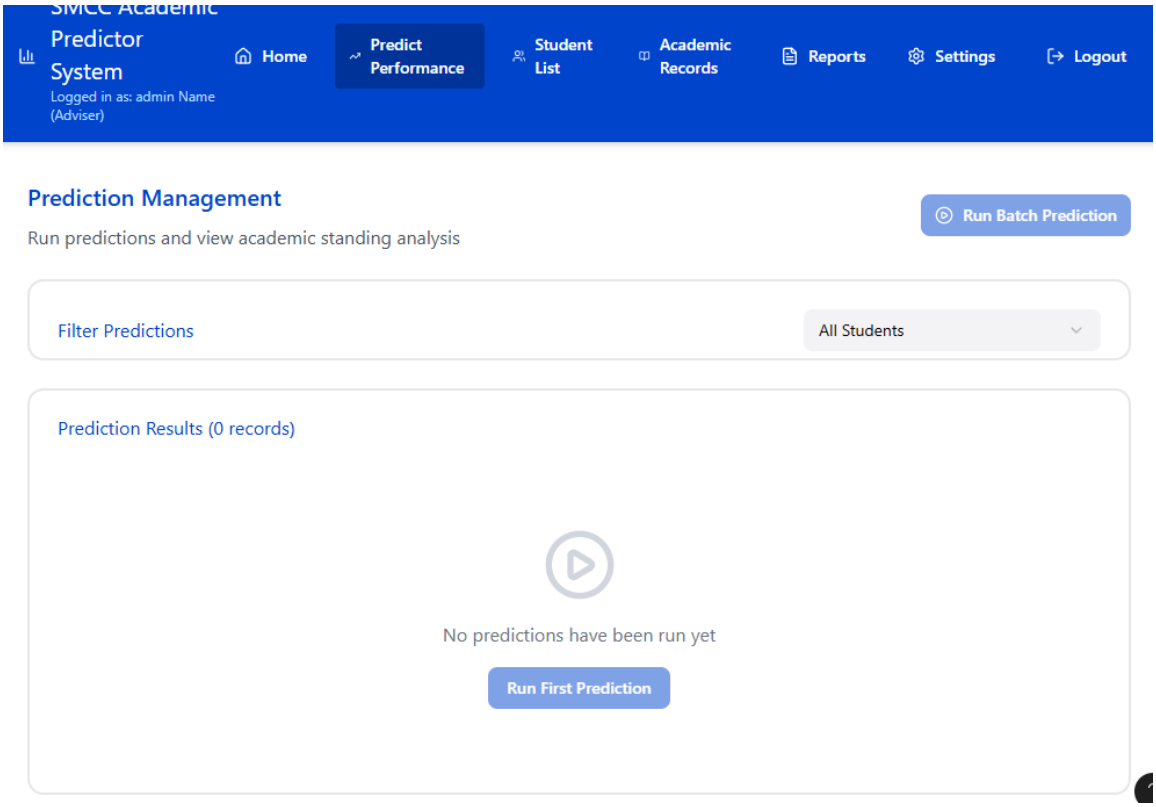


Figure 10: Prediction Management

Figure 10 shows how users interact with the prediction system to manage and execute academic standing predictions. Users can run batch predictions, filter by students, and review prediction results alongside SHAP explanations. The system ensures that both outcomes and reasoning are displayed, supporting transparency and model interpretability. Conditional branches facilitate retries on failures, while distinct interface lifelines support streamlined prediction tasks and XAI visualization.

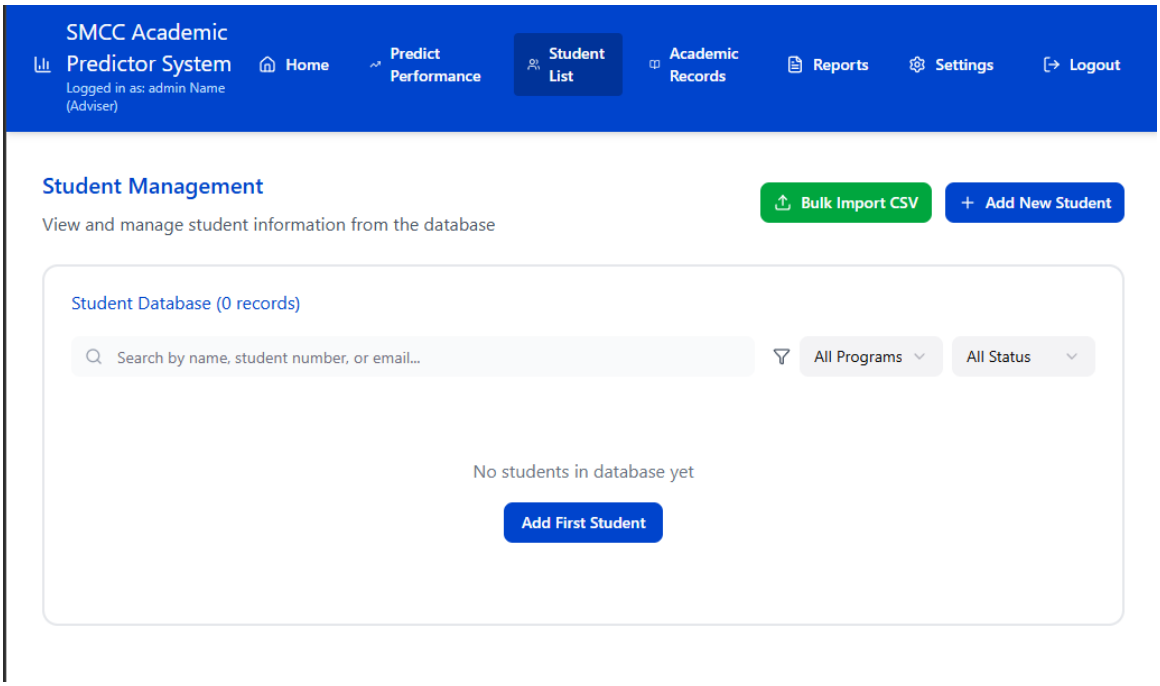


Figure 10: Student Management

This figure shows how users interact with the system to manage the student database, including adding new students and bulk importing via CSV. The management module supports search, filters, and status updates, integrating directly with the centralized database for validated storage. The workflow includes error handling and retry logic, and connects to prediction modules where SHAP explanations ensure interpretability for each student's predicted standing.

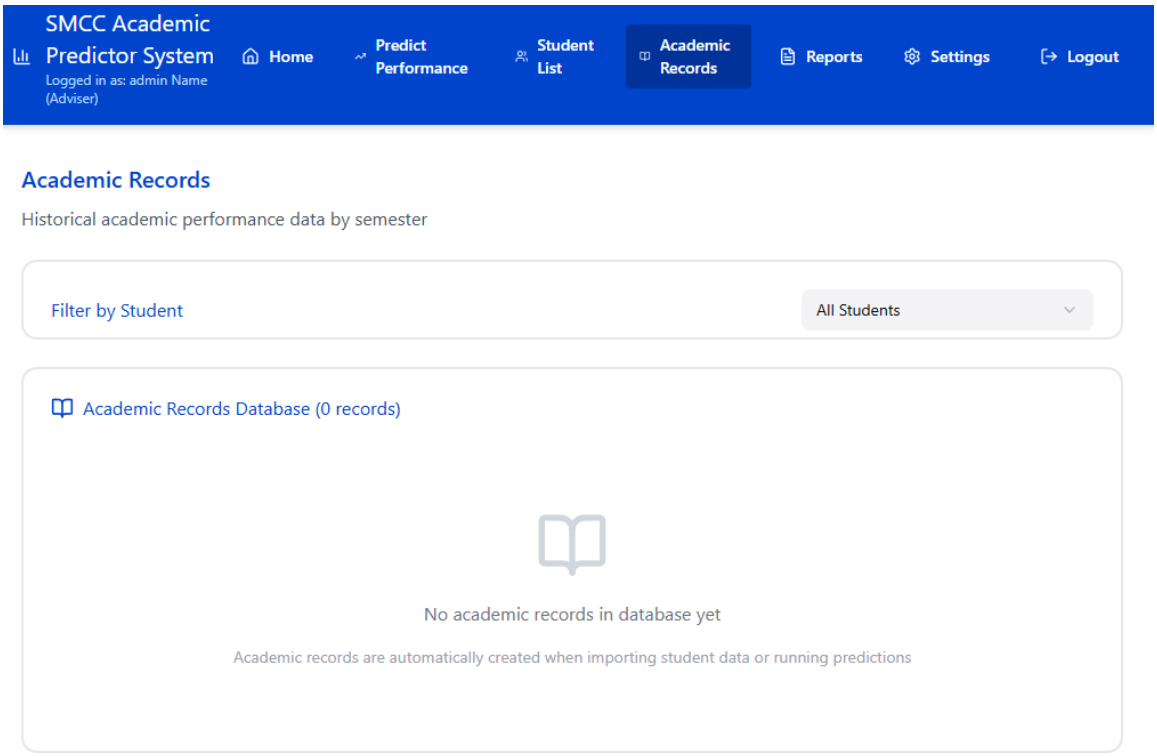


Figure 12: Academic Records

This figure shows how users interact with the prediction system to filter, review, and manage historical academic records by student and semester. Academic records are synchronized with the centralized database and automatically updated by data import and prediction activity. The interface supports conditional messaging for empty data states, while integrating with prediction and SHAP explanation workflows to maintain full transparency and interpretability of academic analysis.

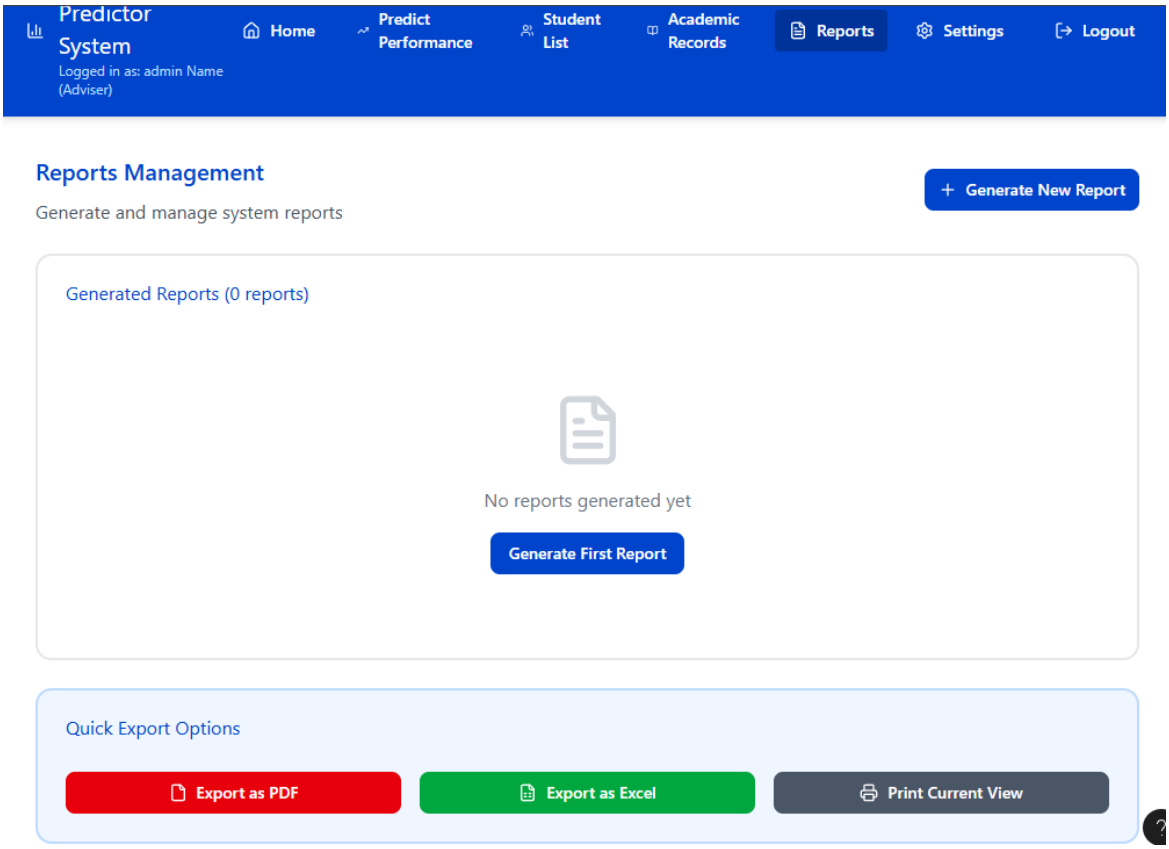


Figure 10: Reports Management

This figure shows how users interact with the prediction system to manage generated reports, export data, and print system views. The reporting interface enables users to create, review, or export results, with the system highlighting error states when no reports exist. Lifeline separation distinguishes reporting, database storage, and user operations, with SHAP explanations integrated into report insights for enhanced transparency and interpretability.

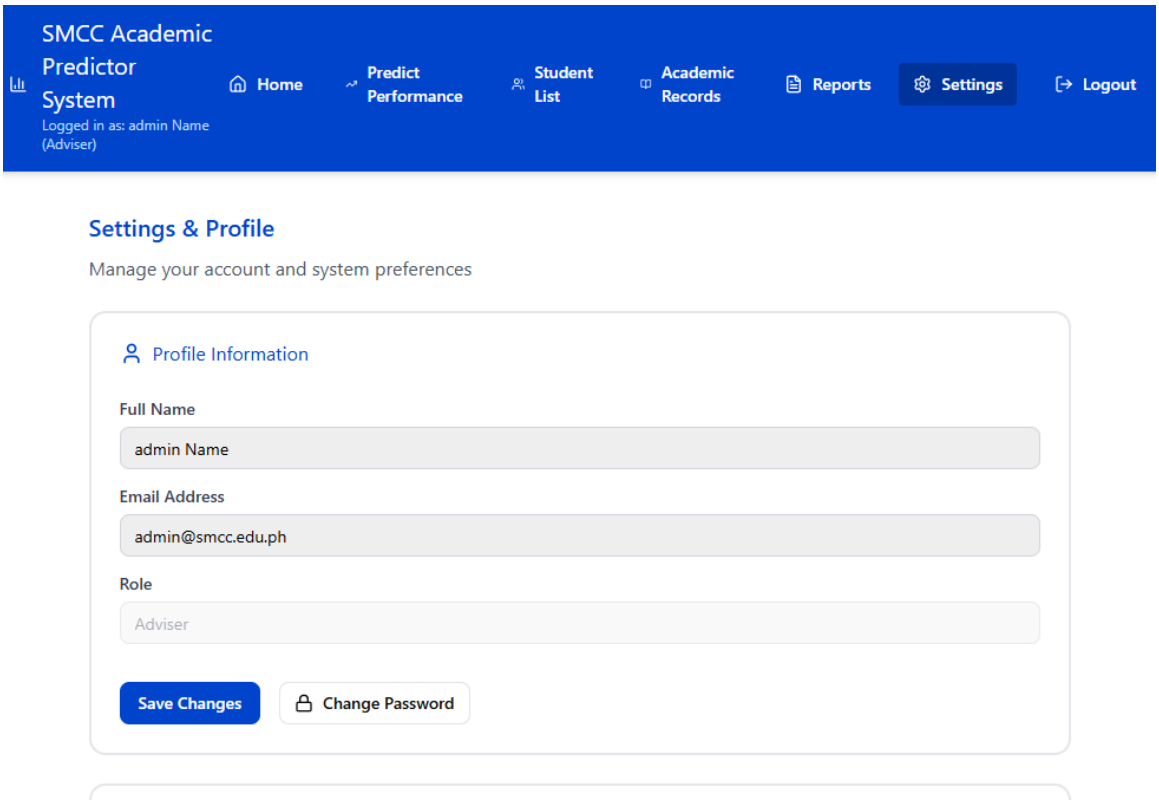


Figure 10: Settings & Profile

This figure shows how users interact with the prediction system to manage their personal account details, update credentials, and review role-based information. Profile management safeguards user authentication and system preferences, with conditional logic handling changes and validation. Distinct system lifelines ensure that user settings are maintained securely and accessed appropriately, supporting traceability and proper data stewardship throughout predictive and explanatory workflows.

2.11 Software Platforms, Development Environments and Tools

Table 1

Software Requirements

Component	Specification	Usage
Front-end		
Jupyter Notebook	JupyterLab (Anaconda)	Interactive Python coding, data visualization, and EDA.
Streamlit (optional)	Python Web UI Framework	Dashboard for presenting prediction results and user interactions.
HTML/CSS/JS (optional)	Web Standards	For custom visualization dashboards if web deployment is required.
Back-end		
Python	3.8+	Develops entire ML pipeline: preprocessing, modeling, automation scripts.
Flask/FastAPI (optional)	Python Web Framework	Provides REST API endpoints for serving predictions in a web environment.
scikit-learn	Python Library	Random Forest algorithm, preprocessing, validation, and evaluation.

pandas	**Python Library**	**Data cleaning, transformation, and merging.**
NumPy	**Python Library**	**Matrix and vector operations for efficient computations.**

SHAP, LIME	**Python Library**	**Model interpretation and feature analysis.**
Matplotlib, Seaborn	**Python Libraries**	**Plots data distributions and evaluation metrics.**
Database/Storage		
MySQL/PostgreSQL	**RDBMS**	**Stores academic, demographic, and LMS records, supports extraction for analysis.**
Excel/CSV	**File Format/Tool**	**Data transfers, backup, and initial cleaning.**
Version Control/Cloud		
Git, GitHub	**Version Control**	**Tracks code development and team collaboration.**
Google Colab/Kaggle	**Cloud Notebooks**	**Cloud computation, easy sharing and scalable model training.**

2.12 Hardware Requirements

Table 2

Hardware Requirements

Components	Specification	Usage
Development PC		
Processor	Intel Core i5 (8th gen) or AMD Ryzen 5	For running data analysis, training models, and development tasks.
RAM	8 GB DDR4	Supports multitasking and fast computation during training and EDA.
Storage	256 GB SSD	Quick access and ample space for large datasets and library dependencies.
Display	1366x768 resolution or higher	Ensures clear visibility of data visualizations and code interfaces.
OS	Windows 10/11, Ubuntu 20.04+, or macOS	Compatibility with necessary ML libraries and tools.
Internet Access	Broadband (10 Mbps+)	Enables data collection, cloud computation, collaboration, and API access.

Peripheral Devices	Mouse, Keyboard, Webcam	Standard input and communication devices for remote collaboration/reporting.

2.13 System Evaluation/Evaluation Procedure

Software Evaluation Using ISO 25010 Standards

The system will be evaluated based on the ISO/IEC 25010 software product quality model, which is the international standard for assessing software quality. ISO 25010 defines eight primary quality characteristics, each with specific sub-characteristics, that will be systematically reviewed during the system evaluation:

- Functional Suitability: Measures how well the system provides functions that meet the stated needs, including completeness, correctness, and appropriateness.
- Reliability: Evaluates stability and error tolerance during operation, including maturity, availability, fault tolerance, and recoverability.
- Performance Efficiency: Assesses system response times, resource utilization, and capacity under specified workloads.
- Usability: Considers the user experience, covering learnability, operability, accessibility, and user interface aesthetics.
- Security: Ensures protection of data and operation against unauthorized access, including confidentiality, integrity, accountability, and non-repudiation.
- Compatibility: Reviews the system's ability to operate in different environments and exchange data with other systems (co-existence and interoperability).

- Maintainability: Examines the ease of modifications, fault diagnosis, and improvements in software (modularity, reusability, analysability, modifiability, testability).
- Portability: Checks adaptability, ease of installation, and replaceability across different environments and platforms.

AI/ML Model Evaluation Metrics

For the predictive analytics component, which uses Random Forest and other machine learning algorithms, system accuracy and robustness will be quantified with the following metrics:

- **Accuracy:** The overall percentage of correctly classified instances among all predictions.
- **Precision**: The proportion of true positive predictions among all positive predictions for a given class.
- **Recall** (Sensitivity): The proportion of actual positives correctly identified by the model.
- **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure for imbalanced datasets.
- **ROC-AUC** (Receiver Operating Characteristic - Area Under the Curve): Evaluates the model's ability to distinguish between classes.
- **Confusion Matrix:** Reports the counts of true positives, false positives, true negatives, and false negatives for all categories, supporting granular performance analysis.

By combining the ISO 25010 framework for general software assessment with specific AI evaluation metrics, the system's usability, performance, security, and prediction validity will be ensured, aligning with international best practices for both software engineering and data-driven modeling.

2.13 Ethical Standards

The development and implementation of the SMCC Academic Predictor System strictly follow established ethical standards to ensure data security, participant protection, and research integrity. The system is designed with respect for the rights, privacy, and wellbeing of all involved parties, maintaining transparency and academic rigor throughout the study.

A. Intellectual Property Rights (IPR)

The institutional data, student records, and information systems used in this study remain the exclusive property of Saint Michael College of Caraga and are accessed only with formal authorization from the institution. The predictive system developed in this research, including its source code and documentation, is created by the student researchers but is governed by the intellectual property policies of SMCC, which may grant the institution shared custodianship over technologies deployed within its environment. The use of institutional names, logos, and platforms in this thesis does not transfer ownership to the researchers and is limited to academic, non-commercial purposes in compliance with official school guidelines.

B. Informed Consent

Student participation in this study is voluntary and is secured through written and/or digital informed-consent forms that clearly explain the purpose of the research, the type of data to be used, and the rights of the participants. Students provide individual consent for the use of their academic and survey data, and they may decline or withdraw at any time without affecting their grades, services, or academic standing. Separately, office-level permission is obtained from the Registrar and other relevant SMCC offices to allow controlled access to institutional records needed for the study; this administrative authorization does not replace or override individual student consent but ensures that data access follows institutional procedures.

C. Data Privacy and Confidentiality

In handling student information, this study complies with Republic Act 10173, the Data Privacy Act of 2012. Access to raw academic, demographic, socioeconomic, and financial records is restricted to the student researchers and the thesis adviser, and, when required, the designated institutional data protection or records officer. All datasets are de-identified before analysis, stored in password-protected devices, and used only for the purposes stated in the informed-consent form. Financial and academic records are treated as sensitive personal information and are not shared outside the research team or reported in individually identifiable form.

D. Voluntary Participation

Participation in this study is strictly voluntary. Students may choose not to participate or may withdraw their participation at any time without penalty. Their decision will not affect their grades, academic standing, access to school services, or any form of evaluation. Any data collected from participants who withdraw will be excluded from further analysis upon their request.

E. Minimization of Harm

This study involves minimal physical risk, but it recognizes potential harms related to data misuse or misinterpretation of predictions. To minimize these risks, all datasets are de-identified before analysis, stored in secure, password-protected systems, and used only for the purposes stated in the consent form. Predictive outputs are treated as decision-support information, not as final judgments about a student's worth or future and are interpreted by qualified academic staff. If any participant feels discomfort regarding the use of their data or their predicted academic standing, they may raise concerns with the researchers or adviser, request clarification, or ask for their identifiable data to be removed from further analysis.

F. Beneficence

This research aims to benefit participants and the institution by enabling earlier identification of students who may be at academic risk and by supporting more timely and targeted interventions. The findings are expected to inform improvements in advising, retention strategies, and resource allocation within SMCC. While individual grades will not be changed by the model, aggregated results and summaries of the study may be shared with interested participants, faculty, and administrators upon request, provided that no student can be personally identified in any report or presentation.

G. Justice and Fair Participant Selection

This study includes first- to third-year students enrolled in BSIT, BSCS, and BLIS programs at Saint Michael College of Caraga who are officially classified as face-to-face students and have complete institutional records (grades, enrollment, and activity logs) for the semesters covered by the dataset. Students from other programs, those with incomplete or missing records, irregular enrollees without consistent term data, and students who did not grant consent for use of their information are excluded from the analysis to protect privacy and maintain data quality.

H. Data Integrity and Accuracy

The predictive models in this study are constrained by the quality and coverage of the available institutional data. Incomplete records, class imbalance between academic-standing categories, and the absence of non-academic variables (such as mental health or employment) may introduce bias into the model's predictions. The reported performance metrics (accuracy, precision, recall, F1-score, ROC-AUC) describe the statistical accuracy of the system's outputs, but they do not guarantee perfect correctness in every individual case; final interpretation and academic decisions remain the responsibility of human advisers who consider additional contextual information beyond the model.

I. Transparency in Reporting

The researchers commit to reporting the results of this study honestly, whether the findings are positive, negative, or inconclusive. All model performances, including cases where algorithms perform poorly or do not meet expectations, will be documented and discussed without selective omission or manipulation of outcomes.

J. Use of Patented or Copyrighted Materials

All ethical and data-protection procedures described in this study were actually implemented during the research, not just "if applicable." The study strictly follows applicable intellectual property laws: any patented or copyrighted technologies, software, or materials are used only with proper licenses or written permissions from the rights holders. No proprietary or commercial datasets or tools are accessed without prior authorization, and all third-party resources are used only for academic, non-commercial purposes in coordination with SMCC.

K. Ethical Considerations for Animal and Human Trials

This study does not involve any animal experimentation. Human participation is limited to anonymized educational records from students who provided informed consent, with their privacy and wellbeing protected throughout.

L. Responsible Use of AI and Related Technologies

All key research decisions, system designs, and final interpretations are made by the human researchers, while AI tools are used only for limited assistance (such as drafting ideas or checking grammar) and are always reviewed by the researchers. Safeguards against bias include cross-checking AI outputs with human judgment and aligning all decisions with RA 10173 and school ethical guidelines to avoid unfair or harmful recommendations.

M. Ethical Approval

This study complies with the Data Privacy Act of 2012 (RA 10173) by using anonymized records, obtaining informed consent, and limiting data access to authorized researchers only. Ethical approval will be sought from the Saint Michael College of Caraga Research, Innovation, and Intellectual Development (RIID) Office, and full deployment of the system will proceed only after formal approval is granted and an official reference number is issued.

2.14 Data Gathering Procedure

Data Gathering Procedure

The researchers will implement a systematic and ethical data gathering procedure to ensure the confidentiality and integrity of all student information. The process will be conducted in the following phases:

Phase 1: Securing Consent and Formal Approval

The first step in the data collection process is to obtain informed consent from the student participants. The researchers will distribute a survey that includes a clear and comprehensive consent form, asking for permission to use their academic, personal, and financial data for the study.

Following the collection of student consent, the researchers will draft formal request letters addressed to the heads of the Registrar, and Admission offices of Saint Michael College of Caraga (SMCC). These letters will specify the exact data required for the study. The approval process will follow a formal chain of command:

The letters will first be reviewed and signed by the group's Thesis Adviser.

Next, they will be endorsed by the Research Leader.

Finally, the letters will be submitted to the heads of the respective offices for final approval.

Phase 2: Data Collection and Handling

Upon approval, the researchers will coordinate with each office to collect the necessary data. The requested information includes:

From the Registrar's Office: All academic records and related factors.

From the Admission's Office: Financial information, including monthly family income, sources of income, and details on any financial aid received.

Resources and indicators data reflecting student access and utilization of academic materials.

Study habits questionnaires administered through structured interviews focusing on resources utilization and study behaviors.

Course load details, specifically the number of units enrolled per semester.

The data will be provided in a combination of physical paper documents and digital Excel files.

Phase 3: Data Security and Confidentiality

To comply with data privacy regulations and ethical research standards, the researchers will immediately implement a strict data security protocol. Upon receiving the data, the researchers will consult with their Data Adviser. All paper documents will be digitized, and the complete dataset will be consolidated into password-protected files. Access to these files will be restricted to the researchers only.

For data preservation and to prevent accidental loss, the secured files will be backed up on a private cloud storage service. The collected data will be used exclusively for research purposes and will be permanently and securely deleted from all local and cloud storage locations upon the completion and successful defense of the thesis

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CHAPTER 4

SOFTWARE DEVELOPMENT AND TESTING

4.1 Development Process

4.2 Testing Process

- Accuracy Test

-Functionality Test

-Performance Test

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CHAPTER 5

SUMMARY, CONCLUSION, RECOMMENDATION,

5.1. Summary of Findings

5.2. Conclusion

5.3.Recommendation/s

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APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo

B - Instrument with Informed Consent

C - Map of the Research Locale *(Include Coordinates)*

D - Curriculum Vitae

E - Narrative C Photo Documentation

F- Ethics Committee Approval

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